Lab 7: Fuzz Testing

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Terminal images rendered from actual run logs; display paths normalized.

1. Original application

```
Windows PowerShell

PS C:\Users\vatsa\SE-Labs\lab7>
python elevator_dispatch.py

==== Running parse_request ====
Enter a request as FROM->TO (e.g. 5->9): 5->9
From and To: (5, 9)

==== Running total_travel ====
Enter floors visited as CSV (e.g. 1,5,3,9): 1,5,3,9
Floors Travelled: 12

==== Running load_percent ====
Enter load as 'X of Y' (e.g. 450 of 630): 450 of 630
Load Percent: 71

Process exit code: 0

Rendered from run logs; paths normalized.
```

```
Windows PowerShell

PS C:\Users\vatsa\SE-Labs\lab7>
python elevator_dispatch.py

==== Running parse_request ====
Enter a request as FROM->TO (e.g. 5->9): bad
Traceback (most recent call last):
  File "C:\Users\vatsa\SE-Labs\lab7\elevator_dispatch.py", line 30, in <module>
    main()
    ~~~~^^
  File "C:\Users\vatsa\SE-Labs\lab7\elevator_dispatch.py", line 16, in main
    result = parse_request(user_input)
  File "C:\Users\vatsa\SE-Labs\lab7\elevator_dispatch.py", line 2, in parse_request
    start, end = text.split('->')
    ^^^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)

Process exit code: 1

Rendered from run logs; paths normalized.
```

2. Fuzz tests before fixing

Hypothesis finds failures in all three functions. Selected traceback excerpts:

```
Windows PowerShell

PS C:\Users\vatsa\SE-Labs\lab7>
python -m pytest test_elevator_dispatch.py -k no_crash -q --tb=short --color=no

FFF                                                                      [100%]
[... traceback excerpts; full output in evidence/03_fuzz_before.txt ...]
    | AttributeError: 'NoneType' object has no attribute 'split'
    | Failing test case: test_parse_request_no_crash(
    |     value=None,  # or any other generated value
    | )
    +-----------------------------------

    | AttributeError: 'NoneType' object has no attribute 'split'
    | Failing test case: test_total_travel_no_crash(
    |     value=None,  # or any other generated value
    | )
    +-----------------------------------

    | AttributeError: 'NoneType' object has no attribute 'split'
    | Failing test case: test_load_percent_no_crash(
    |     value=None,  # or any other generated value
    | )
    +-----------------------------------

========================== short test summary info ==========================
FAILED test_elevator_dispatch.py::test_parse_request_no_crash - ExceptionGrou...
FAILED test_elevator_dispatch.py::test_total_travel_no_crash - ExceptionGroup...
FAILED test_elevator_dispatch.py::test_load_percent_no_crash - ExceptionGroup...
3 failed, 6 deselected in 10.73s

Process exit code: 1

Rendered from run logs; paths normalized.
```

Fix: Validate input types, separators and integers; reject empty fields, negative loads and nonpositive capacity. Invalid input returns None.

3. Fixed application and test results

```
Windows PowerShell

PS C:\Users\vatsa\SE-Labs\lab7>
python elevator_dispatch.py

==== Running parse_request ====
Enter a request as FROM->TO (e.g. 5->9): bad
Invalid input.

==== Running total_travel ====
Enter floors visited as CSV (e.g. 1,5,3,9): 1,,3
Invalid input.

==== Running load_percent ====
Enter load as 'X of Y' (e.g. 450 of 630): 1 of 0
Invalid input.

Process exit code: 0

Rendered from run logs; paths normalized.
```

```
Windows PowerShell

PS C:\Users\vatsa\SE-Labs\lab7>
python elevator_dispatch.py

==== Running parse_request ====
Enter a request as FROM->TO (e.g. 5->9): 5->9
From and To: (5, 9)

==== Running total_travel ====
Enter floors visited as CSV (e.g. 1,5,3,9): 1,5,3,9
Floors Travelled: 12

==== Running load_percent ====
Enter load as 'X of Y' (e.g. 450 of 630): 450 of 630
Load Percent: 71

Process exit code: 0

Rendered from run logs; paths normalized.
```

```
Windows PowerShell

PS C:\Users\vatsa\SE-Labs\lab7>
python -m pytest test_elevator_dispatch.py -q --tb=short --color=no

.........                                                                [100%]
9 passed in 13.41s

Process exit code: 0

Rendered from run logs; paths normalized.
```

Result: All 9 tests pass; valid outputs are preserved.